

Blind symbolic interpretation of CMB β -VAE latent spaces: mutual-information regression rediscovers $A_s e^{-2\tau}$

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Abstract

β -VAE compressions of CMB power spectra learn latents that correlate with physical parameters, but turning a latent into an *equation* usually leaks the answer into the analysis (hand-coded reference combinations, calibration-aware losses, post-hoc latent selection). We present a pre-registered, fully blind alternative: symbolic regression whose inner loss and selection metric are both mutual-information estimators — invariant under any bijection of either variable — applied to every latent of two CMB β -VAE checkpoints (TT-only; TT+low- ℓ EE). No derived parameter combination is computed anywhere in the pipeline; all thresholds were frozen before unblinding, with the amplitude sector as internal positive control. The search blindly recovers the textbook combination $\ln(A_s e^{-2\tau})$; the -2 exponent is read out four independent ways (discovered forms; derivative ratio -1.974 ± 0.018 / -1.993 ± 0.008 ; an answer-agnostic ratio detector, -2.000 ; decoder templates, -1.87) and reappears, unsolicited, in another latent’s residual. All 11 latents receive validated symbolic “cards” — a primary coordinate saturating its blindly selected support ($\eta_s \approx 1$) plus a discovered second coordinate — with 10/11 “primarily interpreted” and one honest failure. Symbolic structure does not terminate at two levels: every stage-2 residual remains structured, robustly to an interaction-aware ansatz. The protocol — freeze, validate on a positive control, unblind, a null for every instrument — is a reusable recipe for interpreting scientific latent spaces.

1 Introduction

Neural compressions of cosmological observables now routinely learn latent spaces whose coordinates *look* physical: a β -VAE trained on CMB temperature power spectra organises its most informative latent along the amplitude direction, and correlation with hand-picked parameter combinations is the standard way this is shown [1]. That last step is where interpretability claims become circular. If the analyst computes $A_s e^{-2\tau}$ and correlates it with a latent, the pipeline contains the answer; if a symbolic fit is scored by mean-squared error against the standard-

ised latent, the search is rewarded for reproducing the encoder’s calibration, not for finding its content.

This paper asks — and answers — a narrower, sharper question: *can the symbolic content of every latent be recovered with no hint of the answer anywhere in the pipeline, under pre-registered decision rules?* Our contributions are methodological first and cosmological second:

1. **A blind search objective.** Symbolic regression (PySR [2]) over *raw* Λ CDM parameters only, scored inside the search loop by a fast Gaussian-mixture mutual-information (MI) estimator and post hoc by an independent held-out MI estimate [3]. Both are invariant under bijections of either variable, so the search can only be rewarded for functional dependence (§2.2).
2. **A pre-registered protocol with falsifiable gates.** Every threshold was frozen before unblinding the non-amplitude latents; the machinery had to first *rediscover* known physics on the amplitude sector — and honestly report what it does not explain — before any discovery claim was allowed (§2.3). Blind-analysis practice of this kind is standard in particle physics [4] but rare in representation interpretation.
3. **Results.** The reionization -2 recovered blindly four ways (§3.1); a validated symbolic card for each of the 11 latents in two checkpoints (§3.2); model-level structure — degeneracy breaking visible before any regression, a split (not duplicated) amplitude sector, and a symbolic hierarchy that does not terminate (§3.3–3.4).

We deliberately do not revisit what is established elsewhere — VAE architecture, training, and emulation accuracy are those of the reproduction of Piras et al. [1], and symbolic regression [5, 6] is used off the shelf; the novelty is the interpretation protocol and what it finds.

2 Method

2.1 Targets, data, and what “blind” means

Two stored β -VAE checkpoints are interpreted *as found*, without retraining: a TT-only convolutional VAE with $L=5$ latents, and a TT+low- ℓ EE dual-encoder VAE with $L=6$ (both $\beta = 3 \times 10^{-4}$, training seed 42), from an independent PyTorch reproduction of Piras et al. [1]. Training data are 500k Latin-hypercube samples of $\theta = (\omega_b, \omega_{\text{cdm}}, H_0, \tau, \ln 10^{10} A_s, n_s)$ mapped to CLASS spectra [7]. The regression target for latent k is the encoder posterior mean $\mu_k(\theta)$ over the 50k-row test split; the cached posterior variances give the stochastic latent Z_k and hence each latent’s finite information ceiling $\text{MI}(Z_k; \theta)$ (§2.4). The test split is tiered once and for all: T0 (5k rows) for the searches, T1 (20k) for calibration and audits, T2 (25k) touched only for confirmatory numbers.

Blindness is enforced structurally, not by good intentions: (i) *inputs are the six raw parameters only* — no derived combination ($A_s e^{-2\tau}$ or otherwise) is computed anywhere in the pipeline; (ii) *both the inner loss and the selection metric are MI estimators*, so no stage rewards matching the latent’s scale or calibration; (iii) *latent roles are assigned before any regression* by a disentanglement audit that measures MI between each latent and each raw parameter — the amplitude latents (TT z_2 , EE z_5) are identified from raw-parameter loadings alone.

2.2 Mutual information as the search objective

PySR searches expressions $f(\theta_S)$ over a support S , with unary operators {exp, log, neg, square} and binary {+, −, ×, ÷}, 200 iterations, 15 populations, maximum complexity 20, on 5 000 rows (4 000 fit / 1 000 validation), five independent seeds per configuration. The inner loss is $\exp[-\widehat{\text{MI}}(f(\theta), \mu_k)]$ with $\widehat{\text{MI}}$ a two-component Gaussian-mixture estimator written in pure Julia (EM on 300-sample sub-batches, milliseconds per call, $\sim 80\times$ faster than calling a Python estimator across the language boundary; ~ 5 min per seed at 16 threads). Selection is post hoc: every Pareto-front equation is re-scored with the full `gmm-mi` estimator [3] on held-out rows.

The key property is shared invariance. For any bijections a, b , $\text{MI}(a(f), b(\mu_k)) = \text{MI}(f, \mu_k)$, so the affine form $A_s(\tau - c)$, the literal $A_s e^{-2\tau}$, and $2\tau - \ln A_s$ carry identical scores, and PySR’s parsimony penalty arbitrates among them. The search is therefore rewarded purely for finding *which function of θ the latent is*, never for reproducing how the encoder happens to scale it. A practical corollary found in a 12-configuration hyperparameter sweep: the top-of-front MI is a capacity dial (it grows with any search-size knob), while the budget-matched readout MI at complexity ≤ 10 is flat across every knob — so we *search big and slice small*, reading results at a one-standard-error

knee of the cumulative Pareto envelope rather than at the front’s argmax (Fig. 2).

2.3 Pre-registration, gates, and controls

All decision rules — sufficiency levels, knee definition, semantic-cluster equivalence, recurrence thresholds, subset minimality, residual-audit and level-set pass criteria, and the final status predicates — were frozen in the roadmap document *before* unblinding any non-amplitude latent. The amplitude sector then served as gate G1, a registered positive control: the frozen instruments had to rediscover the $\{A_s, \tau\}$ support, the -2 exponent, and support saturation $\eta_S \approx 1$, *and* correctly report that the primary coordinate leaves a structured residual (an honesty check — instruments that fabricate sufficiency would fail it). G1 passed on both models. Post-freeze judgement calls were not made silently: five deviations (e.g. the seed tolerance used by recurrence scoring, the additive-hierarchy limit at the TT amplitude latent) are recorded in a register and printed on the affected cards. Every instrument ships a null (Table 4): shuffled targets for the searches, permutation nulls for the residual audits, shuffled residuals for stage two, shuffled-form and wrong-latent controls for the level sets, shuffled probes for the subspace analysis.

2.4 The instrument ladder and the η hierarchy

Table 1 lists the eleven frozen instruments. Three deserve definition here because they separate claims that are usually conflated. For a candidate coordinate f of latent k with blindly selected support S^* :

- $\eta_S = \hat{I}(f)/\hat{I}_{S^*}$ — does f saturate its own support’s MI ceiling?
- $\eta_{\text{plat}} = \hat{I}(f)/\hat{I}_{\text{plat}}$ — what fraction of the SR-accessible mean map does it carry?
- $\hat{\eta}_{\text{post}} = \text{MI}(Z; f)/\text{MI}(Z; \mu)$ — what fraction of what the latent *physically stores*?

A coordinate can honestly score $\eta_S = 1.00$ while $\eta_{\text{plat}} = 0.35$: it is the best one-dimensional summary on its inputs, yet the deterministic mean map $\mu_k(\theta)$ carries much more. That distinction — enforced by the stochastic ceiling $\text{MI}(Z_k; \theta)$, since the mean map can carry detail the sampled latent cannot transmit (residual variance up to $382\times$ the posterior noise here) — is what makes the final statuses precise rather than rhetorical. Semantic recurrence R_{SR} likewise replaces string matching: across-seed forms count as the same discovery only if algebraically identical (canonical simplification), or rank-correlated $|\rho| \geq 0.98$ on fixed anchor rows, or gradient-aligned (cosine distance ≤ 0.05), pooled by union-find with cohesion diagnostics.

Table 1: The instrument ladder. Each row is one frozen instrument; later rows consume earlier rows’ outputs. MI estimates use `gmm-mi` on held-out tiers; every decision statistic is compared against the listed null.

Instrument	Question it answers	Decision statistic	Null / control
Disentanglement audit	which raw parameters does z_k carry?	$\text{MI}(z_k; \theta_j)$ matrix	—
Blind SR (§2.2)	which function of θ is μ_k ?	held-out MI of Pareto forms	shuffled target
Plateau/knee readout	where does the front saturate?	envelope plateau, 1-SE knee c^*	—
Semantic recurrence	does the <i>same function</i> recur across seeds?	R_{SR} via 3 equivalence tests	anti-chaining diagnostics
Blind subset selection	minimal input support S^* ?	all-63-subset MI ceilings $\hat{I}_S$	screen recurrence flags; sham-input dilution control
Calibrated sufficiency	does f_1 account for μ_k ?	η_S ; residual R_{res}^2 audit	permutation null
Intrinsic ceiling	fraction of what z_k stores?	$\hat{\eta}_{\text{post}} = \text{MI}(Z; f) / \text{MI}(Z; \mu)$	DPI sanity
Residual SR (stage 2)	is the residual itself symbolic?	recurrent f_2 ; MI vs. e_1	shuffled residual
Level-set audit	do level sets of (f_1, f_2) pin μ_k ?	$E_{\text{inv}} \leq 0.05$	shuffled-form; wrong-latent
Decoder templates	does the <i>decoder</i> agree?	$\cos(a^*, g_j)$; r_{dec}	template fit R_W^2
Subspace probes	where does the code store f ?	carrier sets; conditional-MI synergy	shuffled probe

3 Results

3.1 The amplitude sector: one exponent, four blind readouts

The TT spectrum constrains A_s and τ above the reionization scale only through $A_s e^{-2\tau}$ [8]. Both checkpoints devote a latent to this direction, and the blind search recovers it. Feeding the search raw (A_s, τ) only, the TT amplitude latent returns $A_s(\tau - 0.598)$ in 5/5 seeds (held-out MI 0.822 nat, cross-seed spread ± 0.025); the EE amplitude latent returns the $A_s(\tau - 0.579)$ family and, in one seed, the literal $249.9 A_s e^{-2\tau} + 0.118$ (1.196 ± 0.001 nat). These *are* the textbook combination: the first-order Taylor expansion of $A_s e^{-2\tau}$ about the prior mean $\bar{\tau}$ predicts the affine form with $c = \bar{\tau} + \frac{1}{2} \approx 0.570$, matching the discovered constants to 1.5–5% (Fig. 1). That TT returns only the affine form is structural, not seed luck: over the prior $\tau \in [0.01, 0.13]$, $e^{-2\tau}$ is 99.9% linear (maximum deviation 0.52%), the two forms tie in MI, and parsimony deterministically prefers complexity 5 over 9. A regex scan over all Pareto-front equations finds the textbook direction in 28/78 (TT) and 37/78 (EE) equations; the literal exponential appears only in EE, whose latent is cleaner.

The -2 exponent itself is then read out four independent ways (Table 2) — including with all six parameters exposed and no latent pre-selected, and including one readout that lives entirely in the observable domain. A fifth, unsolicited appearance: the EE H_0 -latent’s *residual* coordinate (stage 2, §3.4) is literally $A_s e^{-2\tau} / \omega_b^2$ with ratio -2.0000 , found blindly twice — once by the additive stage-2 run and again by the independent interaction-aware rerun.

Both regimes matter. In TT the degeneracy is *present*: the single amplitude latent is forced into the combina-

tion. In TT+low- ℓ EE the reionization bump breaks the degeneracy, the model *could* separate A_s from τ — and its amplitude latent still organises as $A_s e^{-2\tau}$. The agreement of the forced and the emergent arm is the result.

3.2 A validated card for every latent

Extending the frozen protocol from the amplitude control to all 11 latents yields Table 3. Three claims survive every gate:

Every latent has a one-dimensional symbolic primary coordinate that saturates its own support ($\eta_S = 0.99$ – 1.07 against ceilings from a blind $2^6 - 1 = 63$ -subset campaign) and recurs across search seeds ($R_{\text{SR}} \geq 0.8$ everywhere, 8/11 at 1.00). The amplitude latents return $S^* =$ all six parameters as a reproducible superset finding — the all-parameter ceiling genuinely beats $\{A_s, \tau\}$ because shape-sector modulation of the latent is real information. A post-closure, pre-registered rerun of the *complete* 63-support grid at full protocol under a budget-matched readout later resolved the three supports flagged as non-recurrent at the screen tier and bounded input-set search dilution; its crossed outcome is reported with the negative results (§3.5).

Every stage-1 residual is structured ($R_{\text{res}}^2 = 0.96$ – 0.994 against permutation nulls ≈ 0): no latent is a clean one-function channel. Stage-2 search on the residuals (§3.4) gives each latent a discovered second coordinate, lifting $\hat{\eta}_{\text{post}}$ to 0.59–0.96.

One latent fails honestly. EE z_0 has the weakest stage-2 account ($R^2 \approx 0.4$) and its joint level sets miss the frozen invariance bar ($E_{\text{inv}} = 0.060$ – 0.062 vs. 0.05). The blocker is localised — its residual holds more than one coordinate’s worth of structure — and the card says so rather than stretching a predicate.

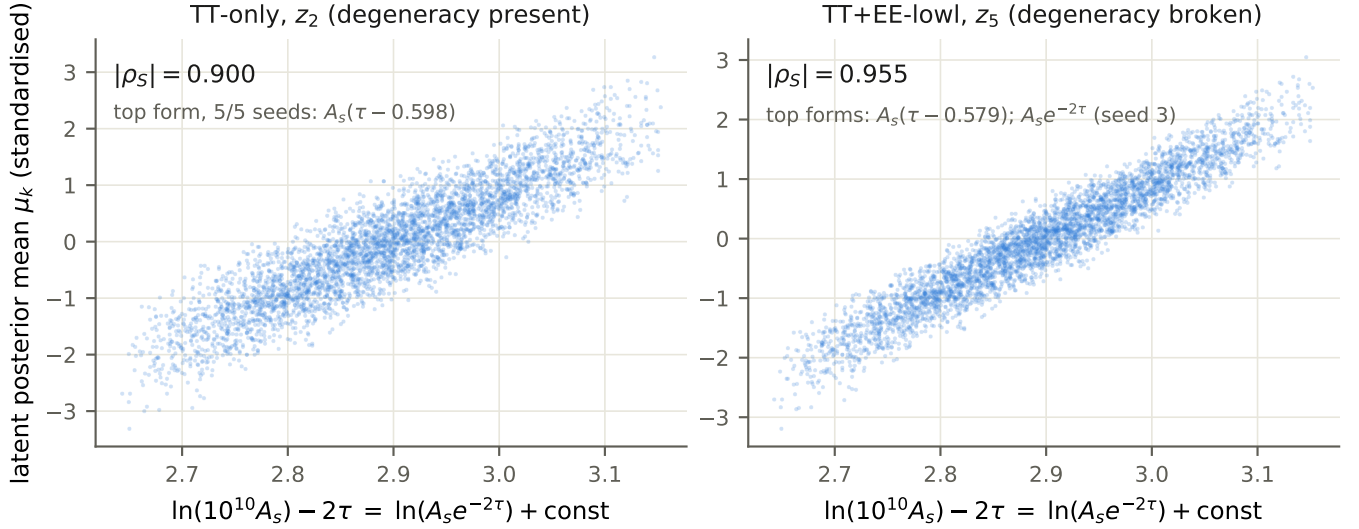


Figure 1: The amplitude latent of each checkpoint against the textbook combination (computed *only* for this figure; the search never sees it). Points: 4000 of the 25k confirmatory-tier rows; ρ_S is the Spearman correlation on the full tier. The search, fed only raw (A_s, τ) , returns the combination’s first-order Taylor form $A_s(\tau - c)$ with c within 1.5–5% of the predicted $\bar{\tau} + \frac{1}{2} \approx 0.570$, and (EE, seed 3) the literal $A_s e^{-2\tau}$.

Table 2: Four independent blind readouts of the reionization exponent (textbook value -2), plus its unsolicited fifth appearance. Each row is a different instrument on different data; none computes the textbook combination.

Readout	Instrument	Value
1. Discovered forms	2-input blind SR, $(A_s, \tau) \rightarrow$ amplitude latent	TT: $A_s(\tau - 0.598)$, 5/5 seeds; EE: $A_s(\tau - 0.579)$ + literal $A_s e^{-2\tau}$; affine c matches Taylor prediction to 1.5–5%
2. Derivative ratio	6-input blind SR, all latents; $r = (\partial_\tau f)/(\partial_{\ln A_s} f)$ on top forms	-1.974 ± 0.018 (TT); -1.993 ± 0.008 (EE)
3. Constant-ratio detector	frozen answer-agnostic ratio field $\rho_{ij} = (\partial f / \partial u_i) / (\partial f / \partial u_j)$, $cv < 5\%$	$(\tau, \ln A_s)$ pair emerges with $r_{\text{raw}} = -2.000$ (TT z_4 , EE z_5 cards)
4. Decoder domain	decoder effect $d_k(\ell)$ decomposed onto data-driven parameter templates	EE reionization bump: $r_{\text{dec}} = a_\tau / a_{\ln A_s} = -1.87$ (TT collinear by mechanism; §3.3)
5. (bonus) Residual reappearance	stage-2 SR on the EE H_0 -latent residual	$f_2 = A_s e^{-2\tau} / \omega_b^2$, ratio -2.0000 ; found independently twice

3.3 Structure of the codes

Three model-level facts, all blind:

Degeneracy breaking has a latent-count fingerprint. The raw-MI audit alone shows one amplitude-sector latent in TT and two in TT+EE — visible before any regression. This is the physics: low- ℓ EE ($\propto A_s \tau^2$) supplies the independent handle on τ .

The EE amplitude sector is split, not duplicated. Sparse-probe and conditional-MI analysis shows z_4 ’s τ -direction is carried by z_4 essentially alone, no redundant latent pair exists in either model (every top-two carrier conditional is synergistic), and conditioning on z_5 raises z_1 ’s information about $A_s e^{-2\tau}$ from 0.013 to 0.473 nat.

Coordinates are latent-anchored, tails distributed. Every f_1 is linearly decodable from its own latent at rank-normal $R^2 = 0.88$ – 0.95 , but no card is axis-aligned; the second coordinates f_2 are genuinely distributed across the code (own-latent $R^2 = 0.01$ – 0.58).

The decoder closes the loop in the observable domain (Fig. 3): per-latent decoder responses $d_k(\ell)$ decompose onto data-driven parameter templates, agree with the symbolic-side sensitivity signatures, and, through the EE reionization bump, deliver the fourth -2 readout. In TT the same construction fails *by mechanism* — τ and $\ln A_s$ are collinear in $\ell \geq 30$ TT template space — which we document rather than patch.

3.4 The hierarchy does not terminate — and the audits agree quantitatively

Stage-2 blind SR on the cached residuals $e_1 = \mu - h(f_1)$ (all six inputs, five seeds) finds a recurrent f_2 for all 11 latents ($R_{\text{SR}} \geq 0.80$; MI against e_1 of 0.40–1.77 nat where shuffled-residual nulls give ≤ 0.074). Combined accounts $h(f_1) + g(f_2)$ reach $R^2(\mu) = 0.955$ – 0.998 — and the frozen stage-2 residual audit *still* fails for all 11: the encoder hierarchy does not terminate at two symbolic levels. This is a

Table 3: Latent cards, both checkpoints (confirmatory-tier values). f_1 : the canonical primary coordinate, discovered blindly on the blindly selected support S^* ; η_S : saturation of the support’s own MI ceiling ($\eta_S > 1$ is possible within estimator tolerance); $\hat{\eta}_{\text{post}}$: fraction of the information the stochastic latent stores, for f_1 alone and with the discovered second coordinate f_2 ; R_{SR} : fraction of seeds recovering the canonical form. Status is assigned by the frozen predicates.

z	role (audit)	f_1 (canonical, blind)	η_S	$\hat{\eta}_{\text{post}}(f_1)$	$\hat{\eta}_{\text{post}}(f_1+f_2)$	R_{SR}	status
<i>TT-only (lcdm_tt_beta3e-4, $L = 5$)</i>							
0	ω_b	n_s/ω_b	1.003	0.531	0.903	1.00	primarily interpreted
1	ω_{cdm}	$H_0^2 n_s^2 \omega_b / (A_s \omega_{\text{cdm}}^2)$	1.000	0.770	0.957	0.80	primarily interpreted
2	amplitude (A_s, τ)	$A_s / \log(H_0(\omega_{\text{cdm}} + \tau))$	1.000	0.347	0.601	1.00	primarily interpreted
3	n_s	$n_s + \omega_{\text{cdm}}$	0.997	0.447	0.838	1.00	primarily interpreted
4	H_0	$A_s H_0^2 \omega_{\text{cdm}} e^{-2\tau}$	1.000	0.371	0.758	0.80	primarily interpreted
<i>TT+EE-lowl (lcdm_tt_ee_lowl, $L = 6$)</i>							
0	ω_{cdm} / early ISW	$\omega_{\text{cdm}} / (H_0 n_s \omega_b)$	1.000	0.756	0.842	1.00	unresolved
1	H_0	$H_0^2 \omega_{\text{cdm}}$	1.000	0.348	0.663	1.00	primarily interpreted
2	ω_b	ω_b / n_s^2	0.995	0.574	0.963	1.00	primarily interpreted
3	n_s	$\log(\omega_b) / (n_s + \omega_{\text{cdm}})$	0.992	0.493	0.864	0.80	primarily interpreted
4	τ (amplitude sector)	$-A_s / (\tau - 0.445)$	1.068	0.842	0.842	1.00	primarily interpreted
5	amplitude (A_s, τ)	$A_s e^{-2\tau}$	1.000	0.289	0.590	1.00	primarily interpreted

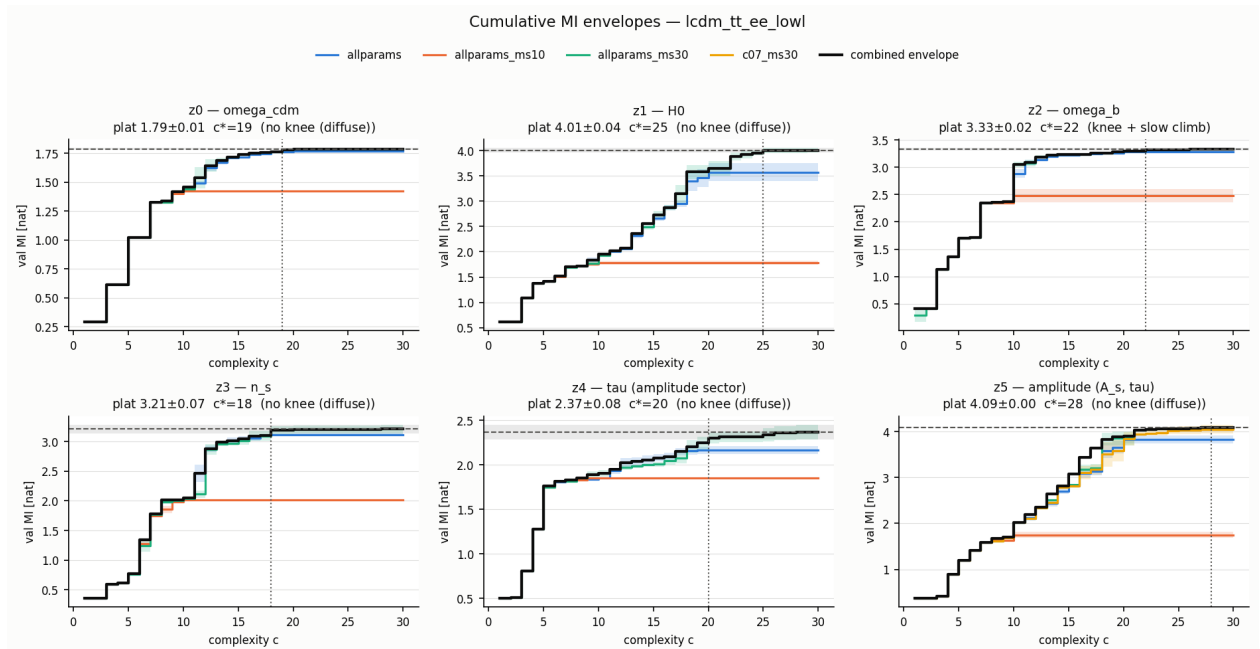


Figure 2: Cumulative Pareto envelopes (validation MI vs. expression complexity), TT+EE-lowl checkpoint; the TT-only analogue is in the repository. Plateaus span 1.79–4.18 nat, one-SE knees sit at complexity 18–28, and 10 of 11 latents’ fronts are diffuse (no sharp knee) — the information is spread over complexity, one face of the non-terminating hierarchy of §3.4. Results are read at the knee, not at the front’s argmax (§2.2).

statement about the representation, not the instruments, and it is robust: a post-closure rerun exposing the stage-1 prediction $\hat{f}_1 = h(f_1)$ as a seventh input (so multiplicative amplitude $\times$ shape interactions are expressible) returns the identical canonical f_2 for 8/11 latents, upgrades EE z_0 ’s f_2 to a unanimous interaction form without rescuing its card, and shows the one registered additive-ansatz limit (TT amplitude latent, deviation D-DoD-z2) is real structure — the search keeps the τ -bearing form even when the interaction channel is available.

The interventional audit agrees with the calibration audit *numerically*, not just in direction. Holding f_1 fixed

and moving everything else, the latent moves: E_{inv} fails the frozen 0.05 rule for all 11 latents — but the failure magnitude is the structured residual itself, $E_{\text{inv}}/(1 - R_{\text{cal}}^2) \in [1.1, 2.6]$ across all latents. Jointly holding (f_1, f_2) restores invariance for 7 of the 8 auditable latents (EE amplitude: $0.226 \rightarrow 0.029$, passing the pre-registered gate on confirmatory data); the three TT latents whose union support is all six parameters have no nuisance direction left to test, and EE z_0 is the one joint failure (§3.2).

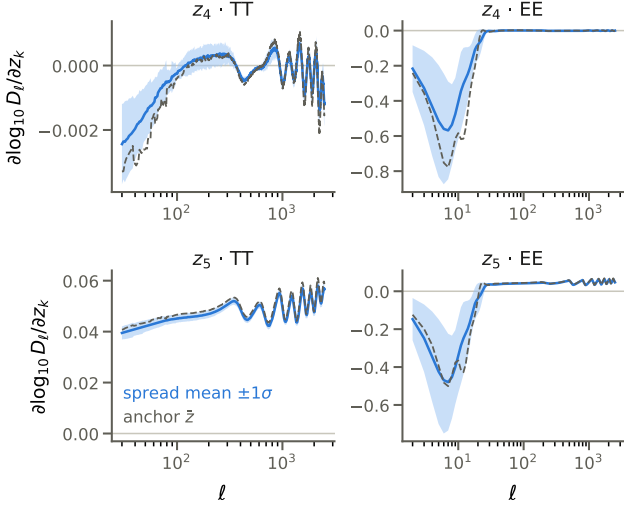


Figure 3: Decoder effect $d_k(\ell) = \partial \log_{10} D_\ell / \partial z_k$ for the amplitude-sector latents of the TT+EE-lowl checkpoint (all 11 latents’ curves are in the repository). Solid: mean $\pm 1\sigma$ over 64 posterior-mean anchors; dashed: the central anchor. The low- ℓ EE reionization bump loads almost exclusively on z_4, z_5 (right column; amplitude template mass 0.79–0.90), which is what makes the observable-domain readout $r_{\text{dec}} = -1.87$ of Table 2 possible; template decompositions achieve $R_W^2 = 0.988$ –1.000 with $\cos(a^*, g_j) = 0.77$ –1.00 against the symbolic-side signatures.

Table 4: Controls (condensed; full table in the repository). Real-signal values in parentheses.

Control	Result
Shuffled-target SR, 2-input amplitude	0.01–0.03 nat (real 0.82/1.20); no textbook structure
Shuffled-target SR, all-params	≤ 0.06 nat on any control front
Stage-1 permutation nulls (R^2)	$p_{97.5} \approx -0.002/0.001$, all 11 latents
Shuffled-residual SR, stage 2	0.009–0.074 nat (real 0.40–1.77)
+ interaction-aware rerun	0.010–0.053 nat
Level sets: shuffled-form E_{inv}	0.81–2.20 (pass bar 0.05)
Level sets: wrong-latent E_{inv}	0.2–2.8 (specificity)
Subspace shuffled probes	$R^2 \approx 0.000$; MI floors ≈ 0.001 nat
Positive control (gate G1)	known answer recovered <i>and</i> its structured residual reported

3.5 Negative results worth keeping

The hyperparameter sweep (12 configurations $\times$ 3 seeds $\times$ 2 models) yields three reusable search-hygiene facts: (i) top-of-front MI measures search capacity, not discovery — maximum expression size alone swings it 1.6 $\rightarrow$ 4.05 nat while the winning composites differ structurally seed to seed; (ii) the budget-matched MI at complexity ≤ 10 is flat across every knob (≈ 1.89 nat TT / 2.02 EE) — the

leading-order discovery is saturated at protocol settings, and the recovered physics is hyperparameter-robust ($r \approx -1.97$ to -2.00 in every configuration); (iii) a small search is not a shortcut to parsimony: the maxsize-10 search’s own front is *worse* at complexity ≤ 10 than the sliced maxsize-20 front (1.60 vs. 1.89 nat TT; 1.69 vs. 2.02 EE).

A fourth fact comes from a post-closure, pre-registered experiment (R-P4): *all* 63 supports rerun at full protocol (3 465 grid cells, five seeds) and read with a budget-matched statistic $M_S^{(c \leq 10)}$ under a frozen select-then-confirm rule, alongside a sham-input control — the all-parameter search rerun with a seventh input that is a permuted copy of a real parameter column, hence provably carrying zero information. The two legs cross. On TT the sham input demonstrably dilutes the search ($\Delta_{\text{sham}} = -0.040 \pm 0.013$ nat pooled over 75 paired runs), yet only one strict subset beats all-six on the held-out tier (z_3 , $+0.090 \pm 0.062$); on EE the sham costs nothing ($+0.014 \pm 0.013$), yet five of six latents confirm a strict-subset advantage — three substantive, $+0.09$ to $+0.13$ nat, the strongest at z_4 where $\{\omega_{\text{cdm}}, \tau, A_s, n_s\}$ beats all-six by $+0.126 \pm 0.047$; the other two pass the frozen $+1$ -SE rule at magnitudes ≤ 0.003 nat and are read as negligible. The sham symbol appears in 0.0% of best-at-budget forms in either model. The practical reading: budget-matched damage from searching on the full input set is real but small ($\lesssim 0.13$ nat), and where subsets win the gain comes from removing specific weakly-coupled true inputs (a distractor effect), not from a smaller search space per se. The three screen-tier support-recurrence flags are thereby resolved (TT z_0 : all-six stands; EE z_3, z_4 : genuine subset advantages); no card status or headline number changes.

4 Limitations and scope

All claims are per-checkpoint. Five search seeds establish *search* stability; stability under VAE retraining (R_{model}) was not run — it requires GPU retraining in the parent repository and remains executable later without touching any frozen threshold. The amplitude family’s recurrence across two different architectures and data regimes, with the latent-count fingerprint matching the physics, is partial cross-model evidence, not a substitute. Further: the symbolic hierarchy is non-terminating at the two levels probed (plus the enriched interaction ansatz); a third level was not attempted. EE z_0 remains unresolved with a localised blocker. The TT decoder readout of the -2 is structurally unavailable (template collinearity), so r_{dec} is EE-only. Estimator caveats are handled by design: decisions ride on η ratios, not raw MI, with bootstrap errors and data-processing-inequality checks throughout.

5 Conclusion

A symbolic-regression pipeline whose only reward is bijection-invariant mutual information, wrapped in a

pre-registered freeze–validate–unblind protocol, recovers the CMB amplitude–reionization combination $\ln(A_s e^{-2\tau})$ from two β -VAE encoders with the -2 exponent read out four independent ways, and returns a falsifiable symbolic card for every latent — including one honest failure and an encoder hierarchy that does not terminate at two symbolic levels. Nothing in the ladder of Table 1 is specific to the CMB: it is a recipe any scientific-VAE interpretation study can run, every instrument shipping its own null.

Reproducibility. All searches (≈ 4600 PySR runs, ≈ 7600 CPU core-hours), consolidated deliverables, frozen thresholds, deviation register, and the per-latent cards are in the repository (`cmb-lcdm-sr`); every number above is regenerable from `experiments/*.json` and is a confirmatory-tier (T2) value.

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